

ECO387C - Fall 2024

Discussion 13

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1 Numerical computation of Lucas tree model

1.1 Setup

The household seeks to solve

$$\max_{\{c_t, s_{t+1}, a_{t+1}\}_{t=0}^{\infty}} \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_t) \right]$$

subject to the budget constraint

$$c_t + \frac{s_{t+1}}{R_t^f} + p_t a_{t+1} = s_t + (d_t + p_t) a_t$$

and no Ponzi game conditions. c_t is the household's consumption. s_t and a_{t+1} are risk-free asset and a stock. $1/R_t^f$ and p_t are the prices of these two assets, respectively. Suppose that the dividends (d_t) follow m -state Markov chain: $d_t \in \{\bar{d}_1, \bar{d}_2, \dots, \bar{d}_m\}$ and the transition probability matrix π^d where the element π_{ij}^d denotes the transition probability from state $\bar{d}_i$ to $\bar{d}_j$.

$$\pi^d = \begin{pmatrix} \pi_{11}^d & \pi_{12}^d & \pi_{13}^d & \cdots & \pi_{1m}^d \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \pi_{m1}^d & \pi_{m2}^d & \pi_{m3}^d & \cdots & \pi_{mm}^d \end{pmatrix}$$

Through the Lagrangian approach, we can obtain the Euler equations for the risk-free asset and stock:

- Risk-free bond

$$u'(c_t) = \beta \mathbb{E}_t \left[u'(c_{t+1}) R_t^f \right]$$

- Stock:

$$p_t u'(c_t) = \beta \mathbb{E}_t \left[u'(c_{t+1}) (p_{t+1} + d_{t+1}) \right]$$

In equilibrium, the market clearing requires that $c_t = d_t$, $a_t = 1$, $s_t = 0$. The Euler equations are

$$p_t = \mathbb{E}_t \left[\beta \frac{u'(d_{t+1})}{u'(d_t)} (p_{t+1} + d_{t+1}) \right]$$

We illustrate how to numerically obtain the equilibrium stock price p_t

1.2 Method 1 - System of equations

Due to the Markovian property of the dividend process, the equilibrium price of a stock depends only on the current state of the dividend. This means in equilibrium, $p_t(d_t) = p_{t+1}(d_{t+1})$ as long as $d_t = d_{t+1}$. Denote vector $p = (p_1, \dots, p_m)$ the equilibrium stock price in each state of dividend. For each state $i \in \{1, \dots, m\}$, we can re-write the Euler equation as follows

$$p_i = \frac{\beta}{u'(\bar{d}_i)} \sum_j \pi_{ij}^d u'(\bar{d}_j) (\bar{d}_j + p_j)$$

or

$$-\sum_j \pi_{ij}^d u'(\bar{d}_j) \bar{d}_j = (\pi_{ii}^d - \beta^{-1}) u'(\bar{d}_i) p_i + \sum_{j \neq i} \pi_{ij}^d u'(\bar{d}_j) p_j$$

Now letting

$$x = \begin{pmatrix} u'(\bar{d}_1) p_1 \\ \vdots \\ u'(\bar{d}_m) p_m \end{pmatrix}, \quad b = \begin{pmatrix} -\sum_j \pi_{1j}^d u'(\bar{d}_j) \bar{d}_j \\ \vdots \\ -\sum_j \pi_{mj}^d u'(\bar{d}_j) \bar{d}_j \end{pmatrix}$$

we can write the system of equations compactly as

$$b = (\pi^d - I \beta^{-1}) x$$

so that

$$x = (\pi^d - I \beta^{-1})^{-1} b$$

Then, we can recover the prices point-wise using $p_i = \frac{x_i}{u'(\bar{d}_i)}$ for all $i \in \{1, \dots, m\}$.

1.3 Method 2 - Recursive iteration over the Euler equation

Another approach to solving for the equilibrium stock price p_t in each state is to leverage the recursive structure of the Euler equation. Similar to how the Bellman equation maps the next-period value function to the current-period value function, the Euler equation

for the stock price can be interpreted as a functional operator T that maps the next-period stock price to the current-period stock price. To see this, recall the Euler equation for stock:

$$u'(d)p = \mathbb{E}_{d'|d} [\beta u'(d')(p' + d')] \Leftrightarrow u'(d)p = \mathbb{E}_{d'|d} [\beta u'(d')d'] + \beta \mathbb{E}_{d'|d} [u'(d')p']$$

This can be reformulated as:

$$(Tf)(d) = g(d) + \beta \mathbb{E}_{d'|d} [f(d')]$$

where $f(d) \equiv u'(d)p(d)$ and $g(d) \equiv \beta \mathbb{E}_{d'|d} [u'(d')d']$. Here, $g(d)$ is a constant that depends only on the current state d .

In equilibrium, $p = p'$, which means that we are searching for a fixed point of the operator T . It is straightforward that this operator satisfies the Blackwell sufficient conditions and is a contraction mapping. Consequently, similar to the value function iteration, we can start with an initial guess for the price vector (or more precisely, $f(d)$), and iterate over operator T until convergence to find the fixed point.

Blackwell sufficient conditions Let $K \subset \mathbb{R}^n$ and $B(K)$ be the set of bounded functions $V : K \rightarrow \mathbb{R}$. Suppose that the operator $T : B(K) \rightarrow B(K)$ satisfies:

- **(Monotonicity)** If $W(k) \geq V(k)$ for all $k \in K$ where $V, W \in B(K)$, then

$$T(W)(k) \geq T(V)(k)$$

- **(Discounting)** There exists $\gamma \in (0, 1)$, such that

$$T(V + a)(k) \leq T(V)(k) + \gamma a \quad \text{for all } V \in B(K), a \geq 0, \text{ and } k \in K.$$

Then T is a contraction with modulus γ .

We now verify that the functional operator T on the Euler equation for stock, as defined above, satisfies the two conditions.

- **Monotonicity:** Suppose $W(d) \geq V(d)$ for all $d \in K$. It follows that

$$g(d) + \beta \mathbb{E}_{d'|d} [W(d')] \geq g(d) + \beta \mathbb{E}_{d'|d} [V(d')] \Leftrightarrow (TW)(d) \geq (TV)(d)$$

- **Discounting:**

$$T(V + a)(d) = g(d) + \beta \mathbb{E}_{d'|d} [V(d') + a] = g(d) + \beta \mathbb{E}_{d'|d} [V(d')] + \beta a = T(V)(d) + \beta a$$

$T(V + a)(d) \leq T(V)(d) + \gamma a$ is satisfied with $\gamma = \beta$. Thus, T is a contraction with modulus β .